AI System Inventory

Sentinel Nexus — Generative & Predictive AI Components

RW

June 2026

Contents

Purpose

Per NIST AI 600-1 **GV-1.6** (AI system inventory) and **GV-1.6-003** (inventory entry contents), this document enumerates every AI/ML component operating within the Sentinel Nexus authorization boundary, with the governance metadata required to manage each across its lifecycle. It is reviewed quarterly and on any model promotion.

Inventory entries

For each component: identifier, role, base model + provenance, artifact type, quant, access mode, integrity control, and known issues / oversight.

Model A — Network Baseline Engine (Math Tripwire)

Field	Value
Role	Unsupervised network-flow anomaly pre-filter (Layer 1)
Architecture	Bidirectional LSTM Autoencoder (~1 MB), CPU inference
Data	8-D normalized flow feature vectors (`network_tap` SPI)
Artifact / integrity	safetensors weights + per-feature normalization + **SHA-384 manifest**
Access mode	Internal NATS consumer (`nexus.alerts.baseline`); no external exposure
Oversight	Deterministic $\mu+3\sigma$ threshold; deploy-gated by `train_lstm_ae` calibration

Model B — Adversarial Pattern Classifier

Field	Value
Role	Network/C2 adversarial reasoning (Tracks 2 & 4)
Base model	Mistral-Small-3.1-24B (pinned hf id + revision in `model_config.toml`)
Artifact	QLoRA 4-bit adapter / merged weights; DeepSpeed ZeRO-3 trained
Access mode	Internal vLLM (port 8001)
Oversight	`03_eval_network.py` gates (Track 2 ≥98%, Track 4 ≥95%)

Model C — Spatial Endpoint Expert

Field	Value
Role	Endpoint forensic reasoning with multi-head SpatialProjector
Base model	pinned (`model_config.toml`); QLoRA SFT + CoT
Artifact	LoRA adapter + `spatial_projector.safetensors`; **SHA-384** verified at boot
Access mode	Internal HF Transformers inference (port 8000)
Oversight	`03_eval_model.py` regression gauntlet (OS-context, schema, injection, spatial boundary, cross-space, **SIEM-pivot validity**); 99% deploy floor

Model D — SOAR Critic / Blast-Radius Evaluator

Field	Value
Role	Final autonomous decision gate before containment (DisruptionIndex)
Base model	pinned; DPO/IPO alignment
Artifact	LoRA adapter; **fails CLOSED** (server unreachable → `manual_review_required`)
Access mode	Internal vLLM (port 8002), 3-token decision space
Oversight	`03_eval_critic.py` 4-phase eval (DPO + governance + P/R/F1 + k-fold)

Agentic LLM Hunter Swarm (Model-orchestrating system)

Field	Value
Role	Layer-3 LangGraph DAG: supervisor + 4 experts + adversarial review board + response
Models used	Sovereign failover chain (internal vLLM/Ollama → optional frontier)
Controls	Canary leak tripwire, cognitive sanitizer, read-only tool sandboxes, RBAC tool kits, review-board grounding, HitL circuit breaker, RAG-memory immunity with TTL

Field	Value
Access mode	NATS-triggered; governed SOAR dispatch only — never touches live endpoints except via SOAR / Det Chamber

Frontier models (optional, deployer role)

Field	Value
Providers	Anthropic (primary), Azure OpenAI (corporate) — **sovereign-by-default, off unless enabled + keyed**
Version pinning	**Enforced** — boot refuses a floating alias (`*-latest`) unless `NEXUS_ALLOW_FLOATING_FRONTIER=1` (NIST MP-4.1-007, control NC-3)
Data egress	Cognitive sanitizer DLP scrub on outbound; internal context must not egress (air-gap posture)
Oversight	Provider model cards + SLAs to be captured (POA&M)

Required inventory metadata (GV-1.6-003) — coverage

Item	Where maintained
Data provenance (source, versioning, integrity)	`model_config.toml`, SHA-384 manifests, corpus manifests
Known issues / incident links	`nist_ai_600_1_control_tracker.md` findings; AI incident plan
Human oversight roles	System Security Plan §7
IP / licensed / sensitive-data considerations	Applicability Determinations; training-data curation policy
Underlying foundation models + versions + access modes	This inventory (above)

Supporting / non-generative ML

TurboVec n-gram dedup + hard-negative miner (corpus hygiene), IsolationForest UEBA in the Rust/Python sensor engines, and the Qdrant HNSW vector tripwires — predictive/ML components inside the boundary, governed by the same supply-chain and test controls.